

2024



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Uniting Blockchain and Esports Using Amazon EventBridge with Legendary Play



2

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)

Empowers players

through digital ownership

Increased

security

10,000+

pre-sign-ups

Overview

Creating a great gaming experience starts with prioritizing players. That's why game company [Legendary Play](#) wanted to offer players ownership of their in-game items and characters.

Using Amazon Web Services (AWS), Legendary Play built a new game incorporating blockchain technology and nonfungible tokens (NFTs) so that players could own and sell the results of their in-game efforts. Legendary Play developed the game securely and achieved scalability and cost efficiency on AWS services such as [Amazon EventBridge](#), which developers use to build event-driven applications, existing systems, or software-as-a-service applications at scale across AWS.



Opportunity | Using AWS Services to Incorporate the Blockchain into Mobile Game Esports Heroes for Legendary Play

Founded in 2017, Legendary Play is a self-publishing mobile game developer dedicated to building and operating mobile games for esports fans and gaming enthusiasts. The company's first game, RIVALS Esports MOBA Manager, has achieved more than 1.5 million downloads. Its recent game, Esports Heroes, is one of the first idle role-playing games—which require minimal player interaction for extended gameplay—in the world of professional esports.

In Esports Heroes, players embody pro esports athletes on their path to stardom, join teams with other players, and make their way through tournaments. Players have digital ownership of their characters and, using blockchain technology, can sell, trade, or gift their characters and in-game items outside the game as NFTs, a type of digital identifier stored on a blockchain that cannot be replicated. “When we looked at game experiences that players love, nothing really belonged to them,” says Philipp Karstaedt, founder and CEO at Legendary Play. “A game company could ban an account, and all of it would be gone. Why should the companies be the gatekeepers of that content? We wanted to give a greater sense of ownership to players.”

When Legendary Play began designing Esports Heroes in 2021, it hadn't yet integrated its infrastructure with the blockchain. It needed to find a fast and efficient cloud services provider to effectively enter this new market. Its previous game was already on AWS, so the team was familiar with the services. Furthermore, AWS has an existing presence in esports and could provide valuable support. “Before we started building any architecture, we had the AWS team double-check it,” says Benjamin Schug, chief technology officer at Legendary Play. “They had some great ideas and prevented us from going down dead ends.”

Solution | Achieving Cost Efficiency by Scaling Compute Services to Demand

Using Amazon EventBridge, Legendary Play built a system that supported many different services: a character database, a game server, a portal to buy and sell NFTs, and a minting API to create new NFTs. On top of all that, Legendary Play has a blockchain service that consumes all the events in the system to update the chain. “We wanted to use the blockchain without necessarily becoming experts in that technology,” says Karstaedt. “AWS successfully abstracts the underlying technology so that people don't need to become experts to use it. Because of that, we could focus on building the environment around the game.”

Legendary Play uses a multiaccount strategy, meaning each of its services exists on its own AWS account within the organization. Using this strategy simplifies operations; Legendary Play gives each team access to only the



functionality it needs. At the same time, this structure increases security. “Security is a big concern because we’re dealing with NFTs that are worth real money,” says Schug. “Using Amazon EventBridge, we can define very fine-grained security rules on each of our services and accounts.”

Esports Heroes also offers several small services, which Legendary Play hosts primarily on serverless infrastructure using [AWS Lambda](#), a serverless, event-driven compute service that runs code for virtually any type of application or backend service without provisioning or managing servers. Because AWS Lambda can automatically scale down to nearly zero, many of Esports Heroes’ services cost only cents per month. Legendary Play also uses AWS Lambda to initiate NFT transfers and minting. Alongside AWS Lambda, the developer runs the game server and minting API on [Amazon Elastic Container Service](#) (Amazon ECS), a service for running highly secure, reliable, and scalable containers.

On AWS, Legendary Play prevented unnecessary spending by seamlessly integrating its backend infrastructure with blockchain technology—all without hiring a blockchain developer. Now, the company can use what it has already built for its next games, saving more time and money.

Legendary Play also gained the opportunity to talk about blockchain technology on an eye-to-eye level with gamers and investors. To better understand its customers, Legendary Play is integrating data it collects from its games using [Amazon DynamoDB](#), a fully managed, serverless, key-value NoSQL database designed to run high-performance applications at any scale. “Using the blockchain, we can understand how people interact with digital ownership, such as how often they use it and how it impacts our business,” says Karstaedt.

Outcome | Creating Meaningful Player Experiences

Already, Legendary Play has secured more than 10,000 sign-ups for free NFT characters to be distributed after the game is launched. It has also sold more than 500 early access OG Heroes NFTs to give early adopters a look at the game before anyone else. As Esports Heroes takes off, the company is planning to incorporate more blockchain opportunities. Specifically, it plans to integrate crypto coins into the game when there is a clear benefit for the players.

“Many games put the financial aspect of cryptocurrencies and the blockchain in the forefront rather than the gaming experience,” says Karstaedt. “For us, the first question is always, Does this serve a meaningful purpose for the player? Using AWS, we focus on the experience that we’re creating for our players rather than the tech behind it.”

About Legendary Play

Founded in 2017, Legendary Play is a game company that brings the passion for esports with the fun of mobile games. Legendary Play’s first game, RIVALS Esports MOBA Manager, has achieved more than 1.5 million downloads, with the prelaunch version of its second title, Esports Heroes, now available on all systems.

AWS Services Used



Amazon EventBridge

Amazon EventBridge makes it easier to build event-driven applications at scale using events generated from your applications, integrated SaaS applications, and AWS services.

[Learn more »](#)

AWS Lambda

AWS Lambda is a compute service that runs your code in response to events and automatically manages the compute resources, making it the fastest way to turn an idea into a modern, production, serverless applications.

[Learn more »](#)

Amazon ECS

Amazon Elastic Container Service (Amazon ECS) is a fully managed container orchestration service that helps you to more efficiently deploy, manage, and scale containerized applications.

[Learn more »](#)

Amazon DynamoDB

Amazon DynamoDB is a serverless, NoSQL database service that enables you to develop modern applications at any scale.

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1 2 3 4 5 6 ... 38

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2

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